SEAT NO. CT-23025

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

THIRD YEAR (COMPUTER SCIENCE/ COMPUTER SCIENCE SPECIALIZATION IN DATA SCIENCE)

FALL SEMESTER EXAMINATIONS 2025

BATCH 2023

Dated: 08-DEC-25

Time: 3 Hours **Max.Marks:60**

Software Engineering– CT-365

Q1: [CLO-2]

a) Identify the type of cohesion in the following scenarios: (3)

i. Suppose there's a module called `manage_patient_records()` that performs different operations depending on what the administrator selects. All the operations (add, delete, update, search) are logically related because they all deal with patient records.

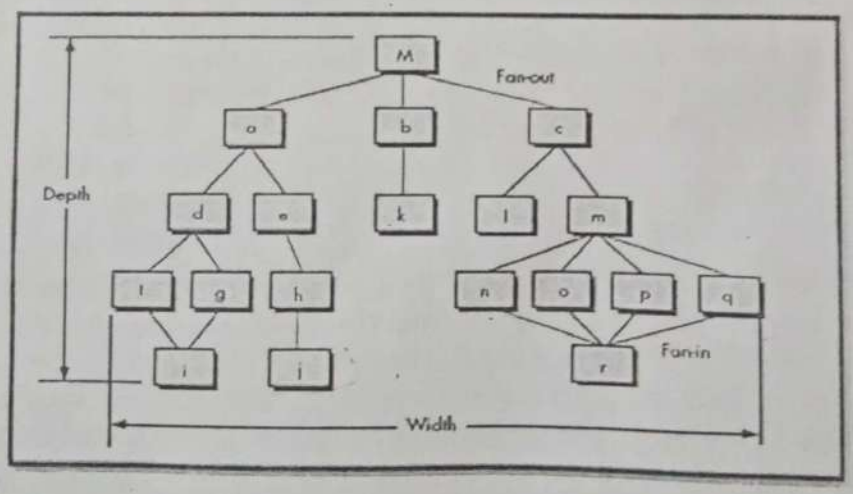
ii. def update_patient_information(patient):
    update_medical_history(patient)
    update_billing_info(patient)
    update_appointment_schedule(patient)
Each operation updates a different part of the patient's record. But all these operations share the same input — the patient's data.

iii. This function patient_bill(patient), sole purpose is to calculate a patient's bill. Every statement contributes to that one goal.

b) Draw the logical and physical data flow diagrams of Patient's appointment system in a hospital. (4)

c) Differentiate between top down and bottom up approach. (3)

d) Compute which of the following modules is highly reusable. (2)



Q2: [CLO-3]

a) An Online Library Management System of the university library wants to automate its book issue/return process. The system includes: (4)
   i. Students logging in to view and borrow books.
   ii. Librarians adding or updating book information.
   iii. Automatic fine calculation for late returns.
   iv. Reports on borrowed books.

The inputs are: Student login, Add new book, Issue book, Return book, the outputs are: Generate fine report, Show issued books, Monthly activity reports. For this purpose the enquiries are Search book and Check fine. Internal files are student file and book file which are used to generate an external file for university database for student verification.
Consider the assigned complexity weights:

Component	Low	Average	High
EI	3	4	6
EO	4	5	7
EQ	3	4	6
ILF	7	10	15
EIF	5	7	10

Calculate the Function Point value for the above scenario assuming the rating of data communication, performance and reusability be 4.

b) Differentiate between size related metrics and function related metrics of software productivity with the help of example. (2)
c) Elaborate the job responsibilities of IT project manager at bank. (2)
d) Explain and List down different types of software maintenance. (4)

Q3: [CLO-3]

a) A company is planning to develop a **500 000 LOC** real-time control system. (8)
The project manager can choose developers from **three different pools**, each with a different mix of analyst capability, application experience, software engineer capability, virtual machine experience and programming language. The Cost Drivers values are provided in table below. Case1: analyst capability and software engineer capability are low and programming language experience is high. Case2: analyst capability and software engineer capability, programming language experience and virtual machine experience are high. Case3: analyst capability, programming language experience, and software engineer capability are high and application experience is low. For each case, compute: effort (E) in PM, development time (D), and the number of persons needed for each project. Then finally decide which developer pool should the manager choose and why?

Cost Drivers	Ratings				
Personnel attributes	Very Low	Low	Nominal	High	Very High
Analyst capability	1.46	1.19	1.00	0.86	0.71
Applications experience	1.29	1.13	1.00	0.91	0.82
Software engineer capability	1.42	1.17	1.00	0.86	0.70
Virtual machine experience	1.21	1.10	1.00	0.90	
Programming language experience	1.14	1.07	1.00	0.95	

b) A system operates for 5 days. During this time, it fails for 5 times, and each failure takes 180 minutes to repair and restore the operation of the system. Calculate the MTBF, MTTR, and the system's availability. (4)

Q4: [CLO-1]

a) Identify whether the example represents a Business Risk, Project Risk, or Technical Risk. (3)

i) A financial trading company deployed an updated software system to automate stock transactions. The software rapidly triggered thousands of unintended transactions, causing massive losses within minutes. The system malfunction caused the company to lose hundreds of millions of dollars in less than an hour.

ii) A national government started a large-scale project to create a unified digital health record system for millions of citizens. The project involved multiple vendors and required coordination across hundreds of hospitals. Over time, requirements kept changing, timelines were extended repeatedly, and the system became increasingly complex. Costs spiraled far beyond the original budget, and after nearly a decade, the project was cancelled without delivering the intended system.

iii) A global mobile manufacturing company once dominated the smartphone market, but its leadership underestimated how quickly users would shift toward full-touch smartphones and modern app ecosystems. Competitors who adapted faster captured the market, causing the company's sales to collapse within a few years.

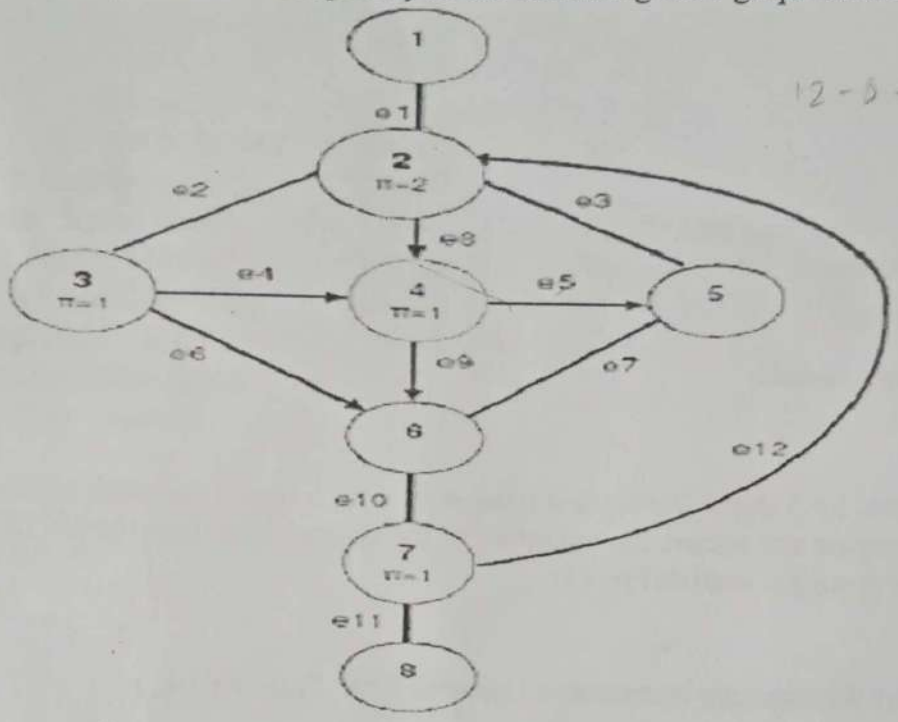
b) Discuss the benefits and limitations of Agile Approach and compare the agile approach with waterfall approach. (4)

c) What are the different roles and responsibilities in Agile and how it is impacted by agile velocity and burn chart? (3)

d) Discuss Risk Control and Mitigation Strategies. (2)

Q5: [CLO-2]

a) Find the Cyclomatic Complexity of the following flow graph of the software code. (2)



12 - 8 + 2 = 6

b) Generate six test cases each of different test category for the Radar Control System. (4)

c) Explain in detail the white-box and the black-box testing. (2)

d) Perform Data-Flow Testing of the given Code and List down its advantages. (4)

```
#include <stdio.h>

int main() {
    int x, y = 5;
    int z = 10;

    if (y > 3)
        x = y + 2;
    else
        x = x + 1;

    x = x * 2;

    if (x < 0)
        printf("Negative\n");

    printf("Final Value: %d\n", x);
    return 0;
}
```